

CSC1005/CSC1022 Database Lab

Lab 4: Nested SQL and Group By Solutions

Goal: Learn aggregate and nested SQL syntax and how to write your basic queries.

This lab is designed to get students familiar with a bit of complex queries. You get to practice group by, aggregate functions and nested queries.

You have finished the lab when you have finished all queries.

If queries specify using a subquery or nested query, you must use a nested query.

Section 1. Basketball (Make default database)

1. Return the year and the average wins of the NBA teams after the 1990 season only when this average is less than 40 (Hint: Group By required)

Ans:

```
select year, avg(won) from teams
      where year > 1990 group by year having avg(won) < 40 ;
```

year	avg(won)
1998	25.000
2011	33.000

2. Find all the basketball coach IDs with more than 900 of wins, order by ascending number of losses (Hint: Group By required)

Ans:

```
select coachID , sum(won), sum(lost) from coaches group by coachID
      having sum(won) >900
      order by sum(lost) ASC;
```

coachID	sum(won)	sum(lost)
auerbre01	938	479
jacksph01	1155	485
.....		
wilkele01	1332	1155

3. Print the players (playerID) of the Miami 1998 team with over 1200 minutes played. Use a nested query in your answer. (Challenge: Can you display the player names as well?)

Ans:

```
select playerID from players_teams
where year = 1998 and minutes > 1200 and tmID in
(Select tmID from teams where name like '%miami%' and year = 1998);
```

playerID
brownpj01
hardati01
majerda01
mournal01
portete01

Challenge:

```
select firstName, lastName, playerID from players
where playerID in
(Select playerID from players_teams
where year = 1998 and minutes > 1200 and tmID in
(Select tmID from teams
where name like '%miami%' and year = 1998));
```

This gives firstName, lastName as well as playerID

Section 2: Sakila (Make Sakila the default database)

4. Return the title and the description of the movie having length of at least 2 hours and containing the word 'wrestler' in the description and 'agent' in the title

Ans:

```
SELECT title, description from film
WHERE length >= 120 and description
Like '%wrestler%' and title like '%agent%'
```

<u>Title</u>	<u>Description</u>
AGENT TRUMAN	A Intrepid Panorama of a Robot And a Boy who must Escape a Sumo Wrestler in Ancient China

5. How many movies last more than 3 hours?

Ans:

```
SELECT count(*) from film where length > 180;
Count : 39
```

6. Select the title and rating of suitable films from the 'Children' category using a nested query. (Hint: you have to use nested queries with multiple tables)

Ans:

```
SELECT title, rating FROM film
WHERE film_id IN
(SELECT film_id FROM film_category
WHERE category_id IN
(SELECT category_id FROM category
WHERE name = 'Children'));
```

<u>title</u>	<u>rating</u>
BACKLASH UNDEFEATED	PG-13
BEAR GRACELAND	R
BENEATH RUSH	NC-17
*****	*****
ZOOLANDER FICTION	R

Section 3. World (Make default database)

7. Count how many cities are located in a European country with more than 90 million inhabitants? Use a nested query.

Ans:

```
SELECT count(name) from city
where CountryCode in
(Select code from country
where Continent = 'Europe' and population > 90000000);
```

Count: 189

8. Return the count and average life expectancy of the countries grouped by continent of origin with an average life expectancy greater than 45 years. (Hint: Group By required)

Ans:

```
SELECT continent, Count(name), avg(LifeExpectancy) from country
group by Continent having avg(LifeExpectancy) > 45;
```

<u>Continent</u>	<u>Count(name)</u>	<u>avg(LifeExpectancy)</u>
North America	37	72.99189
Asia	51	67.44118
Africa	58	52.57193
Europe	46	75.14773
South America	14	70.94615
Oceania	28	69.71500

9. Count the total number of people in all the countries whose official language is Spanish. Use a nested query.

Ans:

```
SELECT sum(population) from country
where code in
(select CountryCode from countrylanguage
where Language = 'Spanish' and IsOfficial = 'T');
```

sum(Population) : 372190700